

SULTAN ISMAIL, Malaysia

WMO: 486790

Lat: 1.6384N

Lon: 103.6660E

Elev: 41

StdP: 100.83

Time Zone: 8.00 (E08)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.8	22.1	20.1	14.8	26.5	20.9	15.6	25.7	7.1	28.9	6.4	29.2	0.7	340	0.285

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	8.0	34.0	27.3	33.2	26.9	33.0	26.8	28.5	32.6	28.1	32.0	27.7	31.5	2.7	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.2	23.1	31.6	27.0	22.8	31.3	26.7	22.3	30.7	92.2	33.1	90.6	31.9	88.6	31.6	31.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.8	5.1	4.4	20.9	35.0	0.9	0.8	20.3	35.6	19.7	36.1	19.2	36.5	18.5	37.1		
(5)			20.2	29.4	0.9	1.2	19.5	30.2	19.0	30.9	18.5	31.6	17.8	32.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.5	26.8	27.3	27.9	28.2	28.1	27.8	27.4	27.3	27.3	27.6	27.4	27.1
	DBStd	1.09	1.23	0.98	1.08	0.94	0.96	1.02	0.98	1.01	0.96	0.93	0.99	1.15
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6393	522	484	554	544	561	533	540	537	521	546	522	529
	CDD18.3	3352	263	251	296	294	303	283	282	279	271	287	272	271
	CDH23.3	29349	2098	2191	2618	2635	2872	2703	2572	2511	2331	2477	2188	2154
	CDH26.7	10713	705	837	1043	1019	1076	1004	939	919	857	902	724	687
Wind	WSAvg	1.5	2.0	2.1	1.6	1.2	1.2	1.2	1.3	1.4	1.3	1.2	1.3	1.7
Precipitation	PrecAvg	2580	236	139	211	245	198	175	169	201	195	233	275	300
	PrecMax	3361	612	402	467	387	297	271	272	329	268	337	382	836
	PrecMin	2075	52	47	75	145	86	112	35	116	61	141	113	133
	PrecStd	349	144	87	108	70	68	50	58	66	49	61	70	166
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.2	33.9	34.2	34.9	34.2	34.0	33.2	33.2	33.5	33.9	33.2	33.2
		MCWB	27.4	26.5	27.0	28.4	27.6	27.5	27.2	26.7	26.9	27.3	27.2	27.4
	2%	DB	32.1	32.9	33.5	33.8	33.2	33.1	32.8	32.8	32.9	33.0	32.2	32.1
		MCWB	26.4	26.2	26.8	27.5	27.1	27.0	26.9	26.5	26.7	26.7	26.7	26.7
	5%	DB	31.2	32.0	32.9	32.9	32.8	32.2	32.0	32.0	32.0	32.1	31.8	31.2
		MCWB	25.9	25.8	26.5	26.9	27.0	26.6	26.4	26.3	26.4	26.3	26.5	26.5
	10%	DB	30.2	31.1	31.9	31.9	31.8	31.2	31.1	31.0	31.1	31.1	30.9	30.2
		MCWB	25.5	25.6	26.1	26.7	26.9	26.6	26.3	26.2	26.4	26.3	26.4	26.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.1	28.0	28.7	29.2	29.0	28.6	28.4	27.8	28.1	28.2	28.1	28.2
		MCDB	32.4	32.5	33.2	33.9	33.0	32.4	32.1	31.6	32.0	32.6	32.3	32.3
	2%	WB	27.2	27.1	27.7	28.3	28.2	27.9	27.6	27.2	27.4	27.5	27.4	27.4
		MCDB	31.2	31.4	32.2	32.4	31.7	31.4	31.2	31.0	31.2	31.5	31.4	31.2
	5%	WB	26.5	26.5	27.2	27.7	27.7	27.4	27.0	26.8	26.9	26.9	26.9	26.7
		MCDB	30.0	30.5	31.4	31.5	31.1	30.9	30.7	30.4	30.6	30.7	30.5	30.2
	10%	WB	26.0	26.0	26.5	27.0	27.2	26.9	26.5	26.4	26.5	26.5	26.4	26.2
		MCDB	29.1	29.8	30.3	30.6	30.5	30.2	29.8	29.7	29.9	29.9	29.5	29.3
Mean Daily Temperature Range	5% DB	MDBR	7.4	8.5	8.7	8.7	8.0	7.7	7.7	7.8	8.0	8.1	7.7	7.1
		MCDBR	8.7	9.5	9.9	9.7	9.1	8.7	8.7	8.9	9.1	9.0	8.8	8.4
		MCWBR	3.7	3.8	3.9	4.1	3.8	3.7	3.7	3.7	3.9	3.9	3.7	3.9
	5% WB	MCDBR	8.0	8.6	8.9	8.9	8.3	8.0	7.9	8.1	8.3	8.3	8.1	7.9
		MCWBR	3.8	3.9	4.2	4.4	4.1	3.9	3.8	3.8	4.0	3.9	3.8	3.9
Clear-Sky Solar Irradiance	taub	0.466	0.479	0.491	0.494	0.485	0.498	0.489	0.530	0.559	0.582	0.480	0.464	
	taud	2.292	2.234	2.209	2.200	2.234	2.198	2.235	2.098	2.015	1.930	2.249	2.303	
	Ebn at Noon	864	858	844	822	808	784	797	782	776	762	845	859	
	Edh at Noon	137	148	151	149	139	142	138	163	181	198	142	134	
All-Sky Solar Radiation	RadAvg	4.48	5.19	5.18	5.06	4.70	4.48	4.44	4.50	4.47	4.49	4.28	4.12	
	RadStd	0.38	0.47	0.47	0.31	0.21	0.31	0.30	0.24	0.32	0.32	0.24	0.31	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (7 neighbors)		+0.67	+0.48	+1.13	+0.60	+1.41	+1.27	N/A	N/A	+249
		+0.36	+0.10	N/A	N/A	+0.18	+0.17	N/A	N/A	+247

Nomenclature: See separate page